

GLADSTONE AP, Australia

WMO: 943810

Lat: 23.8697S

Lon: 151.2214E

Elev: 17

StdP: 101.12

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	10.3	-3.4	2.8	18.2	-0.8	3.5	16.9	8.5	21.9	8.0	21.8	2.7	210	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.3	32.1	23.9	31.0	23.6	30.2	23.2	26.0	29.3	25.4	28.8	24.9	28.3	5.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.2	27.7	24.5	19.5	27.4	23.9	18.8	27.2	80.7	29.5	78.3	28.8	76.2	28.3	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.9	7.9	DB	6.2	35.3	0.8	1.4	5.6	36.4	5.1	37.2	4.6	38.0	4.1	39.1	(4)
(5)			WB	3.3	27.2	1.0	1.0	2.6	27.9	2.0	28.5	1.5	29.1	0.8	29.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.4	26.6	26.5	25.5	23.3	20.4	18.2	17.4	18.0	20.4	22.6	24.4	25.9
	DBStd	3.74	1.43	1.37	1.35	1.46	2.01	2.21	2.02	2.02	1.88	1.81	1.66	1.68
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	109	0	0	0	0	6	28	41	30	3	0	0	0
	CDD10.0	4531	513	461	482	401	321	247	231	247	313	390	432	493
	CDD18.3	1598	255	227	223	151	69	26	13	19	66	132	182	235
	CDH23.3	11814	2445	2139	1818	857	248	44	26	61	274	635	1200	2067
	CDH26.7	2734	699	625	436	103	9	1	1	3	29	85	207	538
Wind	WSAvg	3.9	4.8	4.5	4.2	3.8	3.3	3.1	3.1	3.2	3.5	4.2	4.4	4.5
Precipitation	PrecAvg	929	147	155	101	47	57	37	35	32	24	56	72	133
	PrecMax	1732	640	710	371	250	316	220	170	132	130	277	218	509
	PrecMin	432	6	7	12	0	0	0	0	0	0	3	1	3
	PrecStd	290	129	162	95	51	65	45	43	31	27	55	50	114
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	33.2	32.1	30.0	27.7	25.8	25.5	26.8	29.6	30.7	32.0	33.7
		MCWB	24.9	24.5	23.5	21.7	19.9	18.3	16.5	17.3	20.2	21.2	21.0	24.0
	2%	DB	31.7	31.5	30.5	28.6	26.5	24.5	24.0	24.8	27.2	28.7	29.9	31.5
		MCWB	24.6	24.2	23.4	21.1	19.1	17.8	16.6	17.1	18.6	21.2	22.1	23.6
	5%	DB	30.5	30.6	29.7	27.8	25.7	23.6	23.0	23.7	25.9	27.4	28.7	30.2
		MCWB	23.8	23.8	22.9	20.7	18.6	17.3	16.2	16.5	18.3	20.8	21.6	23.2
	10%	DB	29.7	29.8	28.9	27.1	24.8	22.8	22.1	22.8	24.8	26.4	27.7	29.3
		MCWB	23.3	23.4	22.4	20.4	18.1	16.8	15.9	16.1	17.9	20.1	21.1	22.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.8	26.3	26.2	24.8	22.4	21.2	20.8	20.6	22.3	23.7	25.0	26.7
		MCDB	30.3	30.0	29.1	27.3	25.5	23.2	22.5	23.8	26.1	27.3	28.5	29.4
	2%	WB	26.0	25.7	25.1	23.4	21.2	20.1	19.2	19.6	21.2	22.9	24.0	25.6
		MCDB	29.4	29.1	28.4	26.3	24.4	22.3	21.7	22.5	24.4	26.5	27.5	29.1
	5%	WB	25.4	25.2	24.5	22.4	20.4	19.0	18.4	18.7	20.5	22.1	23.2	24.8
		MCDB	28.7	28.6	27.9	25.8	23.8	22.0	21.0	21.8	23.8	25.7	26.8	28.3
	10%	WB	24.8	24.7	23.8	21.6	19.6	18.1	17.4	17.8	19.8	21.4	22.5	24.2
		MCDB	28.2	28.2	27.4	25.5	23.2	21.4	20.4	21.2	23.2	24.9	26.3	27.9
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.5	6.8	7.8	9.0	9.0	9.8	10.1	9.3	7.7	6.7	6.5
		MCDBR	7.2	7.6	7.7	8.4	9.3	9.4	10.2	10.8	10.5	8.7	7.8	7.5
		MCWBR	2.6	2.7	2.8	3.3	4.1	4.3	4.9	5.0	4.5	3.5	3.0	2.8
	5% WB	MCDBR	6.5	6.7	6.6	7.1	7.8	7.4	8.0	8.6	8.2	7.7	6.9	6.7
		MCWBR	2.9	2.7	2.8	3.2	3.8	3.9	4.7	4.6	3.8	3.5	3.2	3.2
Clear-Sky Solar Irradiance	taub	0.393	0.389	0.375	0.347	0.325	0.317	0.308	0.319	0.356	0.378	0.395	0.395	
	taud	2.521	2.537	2.563	2.629	2.638	2.654	2.659	2.613	2.497	2.474	2.452	2.491	
	Ebn at Noon	951	941	927	913	895	882	905	928	930	941	943	951	
	Edh at Noon	113	109	102	88	80	75	77	88	107	115	120	117	
All-Sky Solar Radiation	RadAvg	6.62	6.15	5.48	4.81	4.08	3.53	4.00	4.86	5.90	6.52	6.88	6.88	
	RadStd	0.47	0.64	0.46	0.40	0.23	0.28	0.30	0.33	0.44	0.46	0.65	0.66	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page